

Continuum Mechanics

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Chapter 06

Isothermal Fluid Mechanics Part 1



Isothermal Fluid Models

Unknowns and Balance Equations



In the Eulerian description of an isothermal continuum, the basic fields have

$$3 + 3 + 1 + 9 = 16$$

scalar components:

$$\boldsymbol{\phi}(\mathbf{X}, t), \qquad \mathbf{v}(\mathbf{x}, t), \qquad \rho(\mathbf{x}, t), \qquad S(\mathbf{x}, t).$$

The kinematic relation, conservation of mass, balance of linear momentum, and balance of angular momentum provide

$$3 + 1 + 3 + 3 = 10$$

independent scalar equations. Constitutive equations must supply the six remaining relations for the symmetric Cauchy stress tensor S.

Constitutive Restrictions



The fluid models in this chapter relate the Cauchy stress S to the mass density ρ and the spatial velocity gradient $\nabla^{\mathrm{x}}\mathbf{v}$.

Any proposed model must satisfy two restrictions:

- it must be consistent with material frame indifference;
- if the material is energetically passive, it must satisfy the mechanical energy inequality.

We first describe ideal and elastic fluids. Ideal fluids are incompressible, whereas elastic fluids are compressible. Later we will discuss Newtonian fluids and the celebrated Navier-Stokes equation.



Ideal Fluids



Definition (Ideal Fluid)

A continuum body with reference configuration B is an *ideal fluid* if the following properties hold:

1. The reference mass density is uniform:

$$\rho_0(\mathbf{X}) = \rho_0 > 0.$$

2. The material is incompressible:

$$\nabla^{\mathrm{x}} \cdot \mathbf{v} = 0.$$

3. The Cauchy stress is spherical:

$$S(\mathbf{x}, t) = -p(\mathbf{x}, t) I.$$

The scalar field p is called the *pressure*.

Reactive Pressure and Surface Traction



The pressure is the multiplier associated with the incompressibility constraint. Thus the stress is purely reactive:

$$S^{(r)} = -pI, \qquad S^{(a)} = O.$$

Because $S = -pI$ is symmetric, balance of angular momentum is automatically satisfied.

If a surface has unit normal $\mathbf{n}$, then its traction is

$$\mathbf{t} = S\mathbf{n} = (-pI)\mathbf{n} = -p\mathbf{n}.$$

Hence an ideal fluid supports normal traction but no shear traction.

Proposition (Uniform Density)

For an ideal fluid,

$$\rho(\mathbf{x}, t) = \rho_0 > 0, \qquad \mathbf{x} \in B_t, \quad t \geq 0.$$

Proof.

The local conservation of mass equation is

$$\dot{\rho} + \rho \nabla^{\mathrm{x}} \cdot \mathbf{v} = 0.$$

Incompressibility gives $\nabla^{\mathrm{x}} \cdot \mathbf{v} = 0$, so

$$\dot{\rho} = 0.$$

In material variables this becomes

$$\frac{\partial}{\partial t} \rho\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) = 0.$$

Proof (Cont.)

Integration in time gives

$$\rho\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) = \rho\big(\boldsymbol{\phi}(\mathbf{X}, 0), 0\big).$$

Using $\boldsymbol{\phi}(\mathbf{X}, 0) = \mathbf{X}$ and the definition of the reference density,

$$\rho\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) = \rho(\mathbf{X}, 0) = \rho_0(\mathbf{X}) = \rho_0.$$

Since every $\mathbf{x} \in B_t$ has the form $\mathbf{x} = \boldsymbol{\phi}(\mathbf{X}, t)$, the result follows. □

Derivation of the Euler Equations

Balance of linear momentum and conservation of mass give

$$\rho_0 \dot{\mathbf{v}} = \nabla^{\times} \cdot (-p\boldsymbol{I}) + \rho_0 \mathbf{b}, \qquad \nabla^{\times} \cdot \mathbf{v} = 0.$$

The material acceleration is

$$\dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\times} \mathbf{v})\mathbf{v}.$$

Also, in components,

$$[\nabla^{\times} \cdot (-p\boldsymbol{I})]_i = (-p\delta_{ij})_{,j} = -p_{,j}\delta_{ij} = -p_{,i},$$

so

$$\nabla^{\times} \cdot (-p\boldsymbol{I}) = -\nabla^{\times} p.$$

Euler Equations for an Ideal Fluid



The spatial velocity and pressure satisfy

$$\rho_0 \left[\frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}} \mathbf{v}) \mathbf{v} \right] = -\nabla^{\mathrm{x}} p + \rho_0 \mathbf{b}, \qquad \mathbf{x} \in B_t, \quad t \geq 0. \tag{6.1}$$

$$\nabla^{\mathrm{x}} \cdot \mathbf{v} = 0,$$

These equations form a coupled system for $\mathbf{v}$ and p. The second equation is both the incompressibility constraint and the reduced conservation of mass equation.

Proposition (Pressure Gauge Freedom)

If $(\mathbf{v}, p)$ *satisfies the Euler equations, then so does*

$$(\mathbf{v}, p + f(t))$$

for every scalar function $f(t)$.

Proof.

The pressure occurs only through its spatial gradient, and

$$\nabla^{\mathrm{x}}\big(p(\mathbf{x}, t) + f(t)\big) = \nabla^{\mathrm{x}} p(\mathbf{x}, t) + \underbrace{\nabla^{\mathrm{x}} f(t)}_{0} = \nabla^{\mathrm{x}} p(\mathbf{x}, t).$$

Thus the Euler equations determine pressure only up to an additive function of time. □

Constraint and Stress Decomposition for an Ideal Fluid

As discussed in Section 5.6, the Cauchy stress in a constrained material decomposes as

$$\mathsf{S}(\mathbf{x}, t) = \mathsf{S}^{(r)}(\mathbf{x}, t) + \mathsf{S}^{(a)}(\mathbf{x}, t).$$

The reactive stress $\mathsf{S}^{(r)}$ enforces the constraint; the active stress $\mathsf{S}^{(a)}$ is specified by a constitutive equation.

For an ideal fluid,

$$\mathsf{S}^{(r)}(\mathbf{x}, t) = -p(\mathbf{x}, t)\mathsf{I}, \qquad \mathsf{S}^{(a)}(\mathbf{x}, t) = \mathsf{O}.$$

Here p is the pressure multiplier associated with incompressibility:

$$\nabla^{\mathrm{x}} \cdot \mathbf{v} = 0, \qquad \text{equivalently} \qquad \det \mathsf{F} = 1,$$

with $\det \mathsf{F}(\mathbf{X}, 0) = 1$ understood in the equivalence.

In the standard material form, the constraint is

$$\gamma\big(\mathsf{F}(\mathbf{X}, t)\big) = 0, \qquad \gamma(\mathsf{F}) = \det \mathsf{F} - 1.$$

Proposition (Frame Indifference of a Constrained Model)

Let $\mathbf{x} = \boldsymbol{\phi}(\mathbf{X}, t)$ *be a motion with current placement* B_t*. For an arbitrary superposed rigid motion, define*

$$\mathbf{x}^* = \boldsymbol{\phi}^*(\mathbf{X}, t) = \mathrm{Q}(t)\boldsymbol{\phi}(\mathbf{X}, t) + \mathbf{c}(t), \qquad \mathrm{Q}(t) \in SO(3).$$

Here $\mathrm{Q}(t)$ *is the rotation and* $\mathbf{c}(t)$ *is the translation. The transformed placement is* B_t^**, and*

$$\mathrm{F}^*(\mathbf{X}, t) = \mathrm{Q}(t)\mathrm{F}(\mathbf{X}, t), \qquad J^*(\mathbf{X}, t) = J(\mathbf{X}, t), \quad J = \det \mathrm{F} > 0.$$

The fields $\mathrm{S}(\mathbf{x}, t)$ *and* $\mathrm{S}^{(a)}(\mathbf{x}, t)$ *denote the total and active Cauchy stresses in* B_t*. Their starred counterparts* $\mathrm{S}^*(\mathbf{x}^*, t)$ *and* $\mathrm{S}^{(a)*}(\mathbf{x}^*, t)$ *denote the corresponding stresses in* B_t^**, evaluated at the corresponding point* $\mathbf{x}^* = \mathrm{Q}(t)\mathbf{x} + \mathbf{c}(t)$*.*

The fields $q(\mathbf{X}, t)$ *and* $q^*(\mathbf{X}, t)$ *are the respective material constraint multipliers, evaluated at the same material particle* $\mathbf{X}$*.*

Proposition (Frame Indifference of a Constrained Model, Cont.)

Let γ be a smooth scalar constraint function. Assume the same constrained stress model is used in both placements:

$$\mathrm{S}(\mathbf{x},t) = \mathrm{S}^{(a)}(\mathbf{x},t) + \frac{q(\mathbf{X},t)}{J(\mathbf{X},t)} D\gamma(\mathrm{F}(\mathbf{X},t))\mathrm{F}(\mathbf{X},t)^{\mathsf{T}},$$

$$\mathrm{S}^{*}(\mathbf{x}^{*},t) = \mathrm{S}^{(a)*}(\mathbf{x}^{*},t) + \frac{q^{*}(\mathbf{X},t)}{J^{*}(\mathbf{X},t)} D\gamma(\mathrm{F}^{*}(\mathbf{X},t))\mathrm{F}^{*}(\mathbf{X},t)^{\mathsf{T}}.$$

Here $\mathbf{x} = \boldsymbol{\phi}(\mathbf{X},t)$, $\mathbf{x}^{*} = \boldsymbol{\phi}^{*}(\mathbf{X},t)$, and $D\gamma$ is the tensor derivative with respect to the deformation gradient.

Suppose that:

1. The constraint function is invariant under every rotation:

$$\gamma(\mathrm{RA}) = \gamma(\mathrm{A}) \quad \text{for all } \mathrm{R} \in SO(3), \quad \det \mathrm{A} > 0.$$

2. For every superposed rigid motion above, the active stress satisfies

$$\mathrm{S}^{(a)*}(\mathbf{x}^{*},t) = \mathrm{Q}(t)\mathrm{S}^{(a)}(\mathbf{x},t)\mathrm{Q}(t)^{\mathsf{T}}.$$

Proposition (Frame Indifference of a Constrained Model, Cont.)

Then the constraint and stress relation are consistent with material frame indifference in the following precise sense.

If the original motion satisfies $\gamma(\mathrm{F}(\mathbf{X}, t)) = 0$, then

$$\gamma(\mathrm{F}^*(\mathbf{X}, t)) = 0.$$

Choosing the multiplier in the transformed placement by

$$q^*(\mathbf{X}, t) = q(\mathbf{X}, t)$$

ensures that its total Cauchy stress satisfies

$$\mathrm{S}^*(\mathbf{x}^*, t) = \mathrm{Q}(t)\mathrm{S}(\mathbf{x}, t)\mathrm{Q}(t)^{\mathsf{T}}, \qquad \mathbf{x}^* = \mathrm{Q}(t)\mathbf{x} + \mathbf{c}(t).$$

Conversely, under the stated assumptions, if total and active stresses obey this transformation law and $D\gamma(\mathrm{F}(\mathbf{X}, t)) \neq \mathrm{O}$, then necessarily $q^*(\mathbf{X}, t) = q(\mathbf{X}, t)$.

Proof.

Exercise. □

Interpreting the Constrained-Model Criterion

Axiom 5.24 requires the *total Cauchy stress* to be frame indifferent. The constraint enters through the reactive stress that enforces it.

The proposition supplies a sufficient consistency criterion for the mechanical constraint and stress relation. It does not assert that arbitrary choices of q^* give a frame-indifferent stress; the scalar transformation $q^* = q$ is the compatible choice.

For incompressibility,

$$\gamma(\mathsf{F}) = J - 1, \qquad D\gamma(\mathsf{F}) = J\mathsf{F}^{-\mathsf{T}},$$

and hence

$$\mathsf{S}^{(r)} = \frac{q}{J}\left(J\mathsf{F}^{-\mathsf{T}}\right)\mathsf{F}^{\mathsf{T}} = q\mathsf{I} = -p\mathsf{I}.$$

Thus the pressure is a frame-indifferent scalar: $p^*(\mathbf{x}^*, t) = p(\mathbf{x}, t)$.

Proposition (Frame Indifference of the Ideal-Fluid Model)

The ideal-fluid constraint $\gamma(\mathsf{F}) = \det \mathsf{F} - 1$ *and active stress* $\mathsf{S}^{(a)} = \mathsf{O}$ *are frame indifferent. Consequently, the ideal-fluid model is consistent with material frame indifference.*

Proof.

Under a superposed rigid motion,

$$\mathbf{x}^* = \mathsf{Q}(t)\mathbf{x} + \mathbf{c}(t), \qquad \mathsf{F}^* = \mathsf{Q}\mathsf{F}, \qquad \det \mathsf{Q} = 1.$$

The constraint satisfies

$$\begin{aligned} \gamma(\mathsf{F}^*) &= \det(\mathsf{Q}\mathsf{F}) - 1 = \det \mathsf{Q} \det \mathsf{F} - 1 \\ &= \det \mathsf{F} - 1 = \gamma(\mathsf{F}). \end{aligned}$$

The active stress vanishes in every frame. Hence

$$\mathsf{S}^{(a)*} = \mathsf{O} = \mathsf{Q}\mathsf{O}\mathsf{Q}^{\mathsf{T}} = \mathsf{Q}\mathsf{S}^{(a)}\mathsf{Q}^{\mathsf{T}}.$$

The preceding constrained-model proposition now gives the conclusion. □

Proposition (Mechanical Work in an Ideal Fluid)

For every ideal-fluid motion,

$$S : L = 0.$$

Consequently, the model satisfies the mechanical energy inequality (Result 5.29) for every isothermal process, whether or not the process is closed.

Proof.

Using $S = -pI$ and $L = \mathbf{sym}(\nabla^{\mathrm{x}}\mathbf{v})$,

$$\begin{aligned} S : L &= -pI : \mathbf{sym}(\nabla^{\mathrm{x}}\mathbf{v}) \\ &= -p\,\mathbf{tr}\big(\mathbf{sym}(\nabla^{\mathrm{x}}\mathbf{v})\big) \\ &= -p\,\mathbf{tr}(\nabla^{\mathrm{x}}\mathbf{v}) \\ &= -p\nabla^{\mathrm{x}} \cdot \mathbf{v} = 0. \end{aligned}$$

Proof (Cont.)

In material variables,

$$S_m : L_m = 0.$$

Thus, for every $\mathbf{X} \in B$ and every interval $[t_0, t_1]$,

$$\int_{t_0}^{t_1} \det F(\mathbf{X}, t)\, S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t)\, \mathrm{d}t = 0.$$

In particular, the integral is nonnegative, as required by the mechanical energy inequality. □



Euler Equations and Fluid Motion

A Fixed-Domain Initial-Boundary Value Problem

Let the fluid occupy a fixed region D. Given $\rho_0 > 0$, a body force $\mathbf{b}$, and an initial velocity $\mathbf{v}_0$, find

$$\mathbf{v} : D \times [0, T] \to \mathcal{V}, \qquad p : D \times [0, T] \to \mathbb{R}$$

such that

$$\left.\begin{aligned} \rho_0 \left[\frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}} \mathbf{v}) \mathbf{v} \right] &= -\nabla^{\mathrm{x}} p + \rho_0 \mathbf{b} && \text{in } D \times [0, T], \\ \nabla^{\mathrm{x}} \cdot \mathbf{v} &= 0 && \text{in } D \times [0, T], \\ \mathbf{v} \cdot \mathbf{n} &= 0 && \text{on } \partial D \times [0, T], \\ \mathbf{v}(\cdot, 0) &= \mathbf{v}_0 && \text{in } D. \end{aligned}\right\} \tag{6.2}$$

Compatibility and Interpretation of (6.2)



For compatibility, the initial field must satisfy

$$\nabla^{\mathrm{x}} \cdot \mathbf{v}_0 = 0 \quad \text{in } D, \qquad \mathbf{v}_0 \cdot \mathbf{n} = 0 \quad \text{on } \partial D.$$

The boundary condition prevents fluid from crossing the fixed impermeable wall.

For bounded D, the velocity is unique under suitable regularity assumptions, while pressure retains the gauge freedom $p \mapsto p + f(t)$. Global smooth existence in three dimensions is not known in general.

Recovering the Motion Map

When $B_t = D$ is known, equations (6.2) determine $(\mathbf{v}, p)$ without first determining the motion map. Afterward, $\boldsymbol{\phi}$ is recovered from

$$\left.\begin{aligned} \frac{\partial \boldsymbol{\phi}}{\partial t}(\mathbf{X}, t) &= \mathbf{v}\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big), \\ \boldsymbol{\phi}(\mathbf{X}, 0) &= \mathbf{X}, \end{aligned}\right\} \qquad \mathbf{X} \in D, \quad t \in [0, T]. \tag{6.3}$$

For each fixed material point $\mathbf{X}$, this is an ordinary differential equation for its current position.



Definition (Pathlines and Streamlines)

A *pathline* is the trajectory

$$t \longmapsto \boldsymbol{\phi}(\mathbf{X}, t)$$

of a fixed material particle $\mathbf{X}$.

For fixed time t, a *streamline* is an integral curve

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}s} = \mathbf{v}(\mathbf{y}(s), t)$$

of the spatial velocity field.

Pathlines give a Lagrangian description. Streamlines give an instantaneous Eulerian portrait. They coincide when $\mathbf{v}$ is steady.

Example (Following Dye in a Fluid)

Imagine a tiny drop of dye that moves with the fluid, with diffusion neglected. Tracking the *same drop* as time passes traces a *pathline*: the actual trajectory of that fluid particle.

A *streamline* instead describes the flow at one fixed instant. Its tangent at each point gives the velocity direction there at that instant. It is a snapshot of directions, rather than the history of one particle.

If dye is *continuously injected at a fixed point*, the visible trail is a *streakline*. It contains different particles that passed through the injection point at different times.

In steady flow, pathlines, streamlines, and streaklines coincide as curves. In time-dependent flow, they can differ because the velocity field changes while the particles move.



Irrotational Motion and Potential Flow

Definition (6.1 (Irrotational Motion))

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion with spatial velocity $\mathbf{v}$ and spin $W = \mathsf{skew}(\nabla^{\mathrm{x}}\mathbf{v})$. The motion is *irrotational* if

$$W(\mathbf{x}, t) = O, \qquad \mathbf{x} \in B_t, \quad t \geq 0,$$

or, equivalently,

$$(\nabla^{\mathrm{x}} \times \mathbf{v})(\mathbf{x}, t) = \mathbf{0}, \qquad \mathbf{x} \in B_t, \quad t \geq 0. \tag{6.4}$$

Geometric Meaning of Irrotational Motion

The vector

$$\mathbf{w} := \nabla^{\mathrm{x}} \times \mathbf{v}$$

is the vorticity. Stokes' Theorem gives

$$\oint_{\partial\Sigma} \mathbf{v} \cdot \mathrm{d}\mathbf{x} = \int_{\Sigma} (\nabla^{\mathrm{x}} \times \mathbf{v}) \cdot \mathbf{n} \, \mathrm{d}A.$$

Thus an irrotational motion has zero circulation around the edge of every admissible open surface Σ.

This agrees with the interpretation of $W = O$: an infinitesimal neighborhood has no local rigid-body spin.



Result 6.2 (Velocity Potential for Irrotational Motion)

Assume B_t is simply connected for every $t \geq 0$. A smooth vector field $\mathbf{v}$ satisfies equation (6.4) if and only if there is a spatial scalar field ϕ such that

$$\mathbf{v}(\mathbf{x}, t) = \nabla^{\mathrm{x}} \phi(\mathbf{x}, t), \qquad \mathbf{x} \in B_t, \quad t \geq 0.$$

The scalar field ϕ is called a *velocity potential*.

Proof of the Easy Direction of Result 6.2.

Suppose $\mathbf{v} = \nabla^{\mathrm{x}}\phi$, where ϕ has continuous second spatial derivatives. Fix t and write

$$v_i = \phi_{,i}, \qquad \phi_{,ij} := \frac{\partial}{\partial x_j}\left(\frac{\partial \phi}{\partial x_i}\right).$$

Equality of mixed partial derivatives gives $\phi_{,ij} = \phi_{,ji}$. Therefore, in a right-handed orthonormal frame,

$$\begin{aligned}\nabla^{\mathrm{x}} \times \mathbf{v} &= (v_{3,2} - v_{2,3})\mathbf{e}_1 + (v_{1,3} - v_{3,1})\mathbf{e}_2 + (v_{2,1} - v_{1,2})\mathbf{e}_3 \\ &= (\phi_{,32} - \phi_{,23})\mathbf{e}_1 + (\phi_{,13} - \phi_{,31})\mathbf{e}_2 + (\phi_{,21} - \phi_{,12})\mathbf{e}_3 \\ &= \mathbf{0}.\end{aligned}$$

Thus the motion is irrotational. This direction does not require simple connectivity of B_t.

For the other direction, see a book on Advanced Calculus. □



Result 6.3 (Acceleration Field for Irrotational Motion)

For any motion,

$$\begin{aligned}\dot{\mathbf{v}} &= \frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 + 2W\mathbf{v} \\ &= \frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 + (\nabla^{\mathrm{x}} \times \mathbf{v}) \times \mathbf{v}.\end{aligned} \tag{6.5}$$

For an irrotational motion,

$$\dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2. \tag{6.6}$$

Proof.

The spatial acceleration is

$$\dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v}. \tag{6.7}$$

By the definition of W,

$$2W\mathbf{v} = \left[\nabla^{\mathrm{x}}\mathbf{v} - (\nabla^{\mathrm{x}}\mathbf{v})^{\mathrm{T}}\right]\mathbf{v}.$$

In components,

$$\begin{aligned}[(\nabla^{\mathrm{x}}\mathbf{v})^{\mathrm{T}}\mathbf{v}]_j &= v_{i,j}v_i \\ &= \frac{1}{2}(v_i v_i)_{,j} \\ &= \left[\frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2\right]_j.\end{aligned}$$

Here j is free and i is dummy.

Proof (Cont.)

Therefore,

$$2W\mathbf{v} = (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v} - \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2. \tag{6.8}$$

Solving for the convective term and substituting into (6.7) gives the first equality in (6.5).

The identity

$$2W\mathbf{v} = (\nabla^{\mathrm{x}} \times \mathbf{v}) \times \mathbf{v}$$

gives the second equality. When the motion is irrotational, $W = O$ and $\nabla^{\mathrm{x}} \times \mathbf{v} = \mathbf{0}$, which gives (6.6). □



Definition (Conservative Body Force)

A spatial body-force field per unit mass $\mathbf{b}$ is *conservative* if there exists a scalar potential β such that

$$\mathbf{b}(\mathbf{x}, t) = -\nabla^{\mathrm{x}}\beta(\mathbf{x}, t).$$

For example, in a uniform gravitational field with vertical coordinate x_3,

$$\beta(\mathbf{x}) = g x_3, \qquad \mathbf{b} = -g\mathbf{e}_3.$$



Result 6.4 (Criterion for Irrotational Motion in Ideal Fluids)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of an ideal fluid with spatial velocity $\mathbf{v}$ and constant mass density ρ_0. Suppose the body force is conservative, $\mathbf{b} = -\nabla^{\mathrm{x}}\beta$. If $\mathbf{v}$ is irrotational at $t = 0$, then it is irrotational for every $t \geq 0$.



Proof.

Let $\mathbf{w} = \nabla^{\mathrm{x}} \times \mathbf{v}$. Taking the curl of the momentum equation and using Result 6.3 gives

$$\rho_0 \left[\frac{\partial \mathbf{w}}{\partial t} + \nabla^{\mathrm{x}} \times (\mathbf{w} \times \mathbf{v}) \right] = \mathbf{0}. \tag{6.9}$$

Indeed, both force terms are gradients:

$$\nabla^{\mathrm{x}} \times \nabla^{\mathrm{x}} p = \mathbf{0}, \qquad \nabla^{\mathrm{x}} \times \nabla^{\mathrm{x}} \beta = \mathbf{0}.$$

The initial condition is

$$\mathbf{w}(\mathbf{x}, 0) = \mathbf{0}.$$

Proof (Cont.)

Expanding the curl in (6.9) gives

$$\frac{\partial \mathbf{w}}{\partial t} + (\nabla^{\times}\mathbf{w})\mathbf{v} - (\nabla^{\times}\mathbf{v})\mathbf{w} + \mathbf{w}(\nabla^{\times}\cdot\mathbf{v}) - \mathbf{v}(\nabla^{\times}\cdot\mathbf{w}) = \mathbf{0}.$$

For an ideal fluid,

$$\nabla^{\times}\cdot\mathbf{v} = 0.$$

Also, $\mathbf{w}$ is a curl, so

$$\nabla^{\times}\cdot\mathbf{w} = 0.$$

Hence

$$\dot{\mathbf{w}} - (\nabla^{\times}\mathbf{v})\mathbf{w} = \mathbf{0}. \tag{6.10}$$

Proof (Cont.)

Let $\mathbf{w}_m(\mathbf{X}, t) = \mathbf{w}(\boldsymbol{\phi}(\mathbf{X}, t), t)$ and define $\boldsymbol{\xi}$ by

$$\boldsymbol{\xi} = F^{-1}\mathbf{w}_m, \qquad \text{equivalently} \qquad \mathbf{w}_m = F\boldsymbol{\xi}. \tag{6.11}$$

Differentiating $\mathbf{w}_m = F\boldsymbol{\xi}$ and solving for $\dot{\boldsymbol{\xi}}$ gives

$$\frac{\partial \boldsymbol{\xi}}{\partial t} = F^{-1}\left[\frac{\partial \mathbf{w}_m}{\partial t} - \frac{\partial F}{\partial t}F^{-1}\mathbf{w}_m\right]. \tag{6.12}$$

Now

$$\frac{\partial \mathbf{w}_m}{\partial t} = (\dot{\mathbf{w}})_m, \qquad \frac{\partial F}{\partial t}F^{-1} = (\nabla^{\times}\mathbf{v})_m,$$

so equation (6.10) yields $\partial\boldsymbol{\xi}/\partial t = \mathbf{0}$.

Proof (Cont.)

Thus

$$\boldsymbol{\xi}(\mathbf{X}, t) = \boldsymbol{\xi}(\mathbf{X}, 0).$$

Because $F(\mathbf{X}, 0) = I$, equation (6.11) gives

$$\boldsymbol{\xi}(\mathbf{X}, 0) = \mathbf{w}_m(\mathbf{X}, 0).$$

Consequently,

$$\mathbf{w}_m(\mathbf{X}, t) = F(\mathbf{X}, t)\mathbf{w}_m(\mathbf{X}, 0).$$

If the initial velocity is irrotational, then $\mathbf{w}_m(\mathbf{X}, 0) = \mathbf{0}$, and hence

$$\mathbf{w}_m(\mathbf{X}, t) = \mathbf{0}$$

for every $\mathbf{X} \in B$ and $t \geq 0$. □



Result 6.5 (Bernoulli's Theorem for Ideal Fluids)

Let an ideal-fluid motion be irrotational with $\mathbf{v} = \nabla^{\times}\phi$, and suppose $\mathbf{b} = -\nabla^{\times}\beta$. Then balance of linear momentum implies

$$\nabla^{\times}\left(\frac{\partial \phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1}p + \beta\right) = \mathbf{0}, \tag{6.13}$$

or, equivalently,

$$\frac{\partial \phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1}p + \beta = f(t). \tag{6.14}$$



Proof.

Using Result 6.3, $\mathbf{v} = \nabla^{\mathrm{x}}\phi$, and $\mathbf{b} = -\nabla^{\mathrm{x}}\beta$, the momentum equation becomes

$$\rho_0 \left[\frac{\partial}{\partial t}(\nabla^{\mathrm{x}}\phi) + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 \right] = -\nabla^{\mathrm{x}} p - \rho_0 \nabla^{\mathrm{x}}\beta.$$

Since $\partial/\partial t$ and ∇^{x} commute,

$$\nabla^{\mathrm{x}} \left(\frac{\partial \phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1} p + \beta \right) = \mathbf{0}.$$

The quantity in parentheses is spatially constant on each connected component, so it equals a function $f(t)$. □

Potential-Flow Boundary-Value Problem



For an irrotational ideal fluid in a fixed region D,

$$\mathbf{v} = \nabla^{\mathrm{x}}\phi.$$

Incompressibility gives Laplace's equation

$$\Delta^{\mathrm{x}}\phi = 0, \qquad \mathbf{x} \in D, \quad t \geq 0, \tag{6.15}$$

and impermeability gives the homogeneous Neumann condition

$$\nabla^{\mathrm{x}}\phi \cdot \mathbf{n} = 0, \qquad \mathbf{x} \in \partial D, \quad t \geq 0. \tag{6.16}$$

After solving for ϕ, equation (6.14) determines p up to the function $f(t)$.



Proposition (Irrotational Flow in a Bounded Fixed Cavity)

If D is bounded and connected, the only solution of (6.15)–(6.16) *is spatially constant. Hence* $\mathbf{v} = \mathbf{0}$.

Proof.

Multiply $\Delta^{\mathrm{x}}\phi = 0$ by ϕ and integrate over D:

$$\begin{aligned} 0 &= \int_D \phi \Delta^{\mathrm{x}} \phi \, \mathrm{d}V \\ &= \int_{\partial D} \phi \nabla^{\mathrm{x}} \phi \cdot \mathbf{n} \, \mathrm{d}A - \int_D |\nabla^{\mathrm{x}} \phi|^2 \, \mathrm{d}V \\ &= - \int_D |\nabla^{\mathrm{x}} \phi|^2 \, \mathrm{d}V. \end{aligned}$$

Therefore $\nabla^{\mathrm{x}} \phi = \mathbf{0}$, so $\mathbf{v} = \mathbf{0}$. □

Example (Uniform Potential Flow)

In $D = \mathbb{E}^3$, let

$$\phi(\mathbf{x}) = U x_1 .$$

Then

$$\Delta^{\times}\phi = 0, \qquad \mathbf{v} = \nabla^{\times}\phi = U\mathbf{e}_1 .$$

With no body force, steady Bernoulli gives

$$\frac{1}{2}U^2 + \rho_0^{-1} p = f ,$$

so p is constant in space. More generally, at two points $\mathbf{x}$ and $\mathbf{x}^*$ in a steady flow,

$$p - p^* = \frac{1}{2}\rho_0 \left(|\mathbf{v}^*|^2 - |\mathbf{v}|^2 \right) + \rho_0 (\beta^* - \beta).$$